Naor Larian

ELECTRICAL ENGINEERING STUDENT | FPGA & DIGITAL HARDWARE

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SUMMARY

Final-year B.Sc. Electrical & Electronics Engineering student (Computers & Communications, expected 2027) focused on **FPGA and digital hardware design**. Designed, simulated, implemented and debugged a VHDL system on a Basys 3 (Artix-7) board, integrating an ESP32 over UART with real-time VGA output. Nearly four years of troubleshooting live satellite communication systems add practical, system-level debugging and fault-isolation experience.

FPGA PROJECTS

- **Cyber Gladiators** – Independent FPGA Project 2026
  VHDL · Basys 3 (Artix-7) · ESP32 · UART · VGA · Implemented on hardware
  - Designed a modular **VHDL architecture** for a real-time system: 640 × 480 VGA output, sprite rendering, **FSM-based control**, collision detection and game logic.
  - **Integrated an ESP32 controller over UART** as the external control interface to the FPGA design.
  - **Verified RTL modules in simulation**, then synthesized and implemented the design in Xilinx Vivado.
  - **Demonstrated the complete design on a physical Basys 3 board**, debugging the UART-to-VGA integration on hardware.

- **RGB Median Filter** – Afeka Team Project 2026
  VHDL · FSM · RAM/ROM · RTL Simulation · Testbenches · Simulation-verified
  - Developed internal **VHDL components** and integrated FSM control, data buffers and RAM/ROM handling for a 3 × 3 RGB median filter.
  - Verified module- and top-level behavior through **RTL simulation and testbenches** using image test data.

PROFESSIONAL EXPERIENCE

- **Technical Support Engineer** Dec 2022 – Present
  AMOS-Spacecom · Satellite Operator
  - Troubleshoot **15–25 technical incidents per week** on live satellite communication systems, correlating RF spectrum measurements, modem indicators and link performance data to isolate faults.
  - Remotely support VSAT systems (modem configuration, link verification, service activation, monitoring) and approximately **10–15 CPI** and **10–15 commissioning activities** per month.
  - Investigate RF interference using spectrum analysis and structured fault isolation, coordinating resolution with customers and engineering teams.
  - Develop and refine **technical SOPs** that improve troubleshooting consistency and make critical procedures easier to find.

EDUCATION

Expected 2027
B.Sc. Electrical & Electronics Engineering
Afeka Tel Aviv Academic College of Engineering
Specializations: Computers & Communications

2022
Practical Engineer Diploma, Electrical Engineering
Handessaim Tel Aviv
Graduated with honors

TECHNICAL SKILLS

HDL & Design
VHDL, RTL design, FSM design, modular architecture

Verification & Debug
RTL simulation, testbenches, on-board hardware debugging

Tools & Hardware
Xilinx Vivado, Digilent Basys 3 (Artix-7)

Interfaces & Memory
UART, VGA, ESP32, RAM/ROM

Programming & Lab
C, Python, MATLAB/Simulink (academic); spectrum analyzer (professional)

RELEVANT COURSEWORK

Advanced Hardware Design, Digital Signal Processing, Signals & Systems, Communication Systems

LANGUAGES

Hebrew – Native
English – Professional working proficiency